7

Spot Price Models: Pricing Path Dependent and American Style Options

7.1 Introduction

In this chapter we discuss the pricing of path dependent and American style options for the spot price models presented in chapter 6. For both these classes of instrument it is rare to find closed-form pricing solutions, and so numerical procedures must be used to obtain the prices and risk management parameters. In earlier chapters we noted that skews and smiles in implied volatilities are also important to take into account in the pricing of energy derivatives. The two main ways in which smiles can be introduced into models – stochastic volatility and jumps – typically result in a loss of analytical tractability for futures/forward prices and option prices and again derivative pricing is achieved via the implementation of numerical techniques. The general numerical procedures we discuss which allow the pricing of complex path dependent and American style derivatives (including contracts with a swing component) are Monte Carlo simulation and trinomial trees, including so called implied trees[1]. In describing these techniques we build on the framework introduced earlier in chapter 1.

We firstly describe the application of the numerical techniques to the models assuming constant parameters. We then describe how the parameters, in particular the spot price volatility, can be made time dependent in order to incorporate seasonal patterns.

7.2 Simulation Methods for Energy Spot Price Models

Recall from chapter 1 that the value of an option can be computed as the expectation of the discounted payoff[2] and that, under Monte Carlo simulation, the expectation is evaluated as the average of all outcomes obtained by simulating many paths, i.e.,

$$\hat{C}_t = \frac{1}{M}\sum_{j=1}^{M} C_{t,j} \tag{7.1}$$

[1] See Clewlow and Strickland (1998) for further details on the implementation of Monte Carlo simulation and trinomial trees.

[2] Where the expectation is taken under the appropriate measure.

where $\hat{C}_t$ is the estimate of the value of the option with $C_{t,j}$ the discounted payoff under the j^{th} simulation. In this section we illustrate the technique applied to both single factor and multi-factor models.

7.2.1 Single Factor Simulation

We illustrate single factor simulation by using the Schwartz model for the energy price defined by equation (6.6) to value a European call option maturing at time T on an s-maturity forward contract ($T \leq s$) with strike price K. Valuation of a derivative of this kind requires a two-stage process. Firstly, the underlying energy price is simulated to time T, the maturity of the option. Secondly, the s-maturity futures price is computed at time T and compared with the exercise price of the option to obtain the payoff. In this model interest rates are constant and so the discounting becomes $P(t,T)$ where today is time zero. As we saw in chapter 6, the process for the logarithm of the spot price in the Schwartz model, $x = \ln S$, is given by[3];

$$dx = \alpha(\hat{\mu} - x)dt + \sigma dz \tag{7.2}$$

Equation (7.2) can be discretised by changing the infinitesimals dx, dt, and dz into small changes Δx, Δt, and Δz;

$$\Delta x = \alpha(\hat{\mu} - x)\Delta t + \sigma \Delta z \tag{7.3}$$

The random increment Δz has mean zero and a variance of Δt, and can therefore be simulated by random samples of $\sqrt{\Delta t}\varepsilon$ where ε is a sample from a standard normal distribution. In order to simulate the full path for the energy spot price we divide the time period over which we wish to simulate, say $t \in [0, T]$, into N intervals such that $\Delta t = \frac{T}{N}$. We then generate values of S_t at the end of these intervals $t_i = i\Delta t$, $i = 1, ..., N$, using equation (7.3) as follows;

$$\begin{aligned} S_i &= \exp(x_i) \\ x_i &= x_{i-1} + \alpha(\hat{\mu} - x_{i-1})\Delta t + \sigma\sqrt{\Delta t}\varepsilon_i \end{aligned} \tag{7.4}$$

where $S_i = S_{t_i}$ is the spot price observed at time t_i. An important aspect of the spot price models discussed in chapter 6 is that any required futures or forward price can be evaluated at the option maturity date, T, as an analytical function of the spot price and parameters of the model. Thus for the single factor Schwartz model, at time T, for each simulated path, we can compute the value of the s-maturity futures price, $F(T,s)$, via equation (6.9) and then compute the payoff of the call futures option as $C_T = \max(0, F(T,s) - K)$. To obtain the estimate of the call price we simply take the discounted average of these simulated payoffs

$$\hat{C}_t = P(t,T)\frac{1}{M}\sum_{j=1}^{M}\max(0, F_j(T,s) - K) \tag{7.5}$$

where $F_j(T,s)$ represents the forward price under the j^{th} simulation.

[3] $\hat{\mu}$ is defined in equation (6.7).

S	α	μ	λ	σ	μhat	K	T	s	N
26.90	0.472	2.925	0.000	0.368	2.782	23.20	0.50	1.00	10

dt	ln(S)	P(0,T)	call_value	SE	M
0.05	3.292	0.951	1.616	0.091	1000

t	0	0.05	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
ε		0.708	0.574	-0.203	-0.006	-0.027	-0.013	0.367	0.629	0.434	-1.140
xt	3.292	3.338	3.373	3.342	3.328	3.313	3.300	3.318	3.357	3.379	3.271

ST	F(T,s)	CT
26.34	24.41	1.22

FIGURE 7.1 Monte Carlo Simulation of Schwartz (1997) 1 Factor Model for a European Futures Option

Figure 7.1 illustrates a single simulation. The spot energy price and the process parameters are chosen to match the observed crude oil forward curve on 31 March 2000. The spot price is \$26.90 with $\alpha = 0.472$, $\mu = 2.925$, and $\sigma = 0.368$. We assume $\lambda = 0$ and so $\hat{\mu} = \mu$hat $= 2.782$. We price a 6 month (T = 0.5) at-the-money (strike price equal to current futures price implying F = K = 23.20) option on a futures contract with an original maturity of 1 year (s = 1.0), and assume 10 time steps (N = 10) so Δt (dt) = 0.05. Interest rates are assumed to be constant at 10%.

The simulation of a single path of the spot price is shown in the box, the first row shows the time, the second row shows the samples from the standard normal and the third line the simulated natural logarithm of the energy spot price at each time step. At the maturity of the option the spot price, represented by ST in figure 7.1, is recovered as 26.34 = exp(3.271), the forward price (F(T,s)) evaluated via equation (6.9), is 24.41 and the option maturity condition (CT), evaluated with a strike price of 23.20 is 1.22. The option price is determined by applying equation (7.5) with M simulations. We call this procedure the simple Monte Carlo method. Figure 7.2 shows the convergence of this simple method – we plot the option price evaluated via simulation for increasing number of simulations from 10 up to 1000. The analytical price of this option, calculated via equation (6.12), is 1.615 and is also shown in figure 7.2. The Monte Carlo estimate after 1000 simulations (call_value) is 1.616 with a standard error (SE) of 0.091.

It is well known that in order to get an acceptably accurate estimate of the option price a very large number of simulations have to be performed. Most implementations of Monte Carlo simulation therefore employ variance reduction methods such as antithetic variance reduction, a technique consisting of creating a hypothetical asset which is perfectly negatively correlated with the original asset. Implementation of this technique is very simple and is described in detail in Clewlow and Strickland (1998). For example, consider pricing a European call futures option with strike K. At the same time as we simulate the energy spot price equation (7.4) we also simulate the antithetic variable $\bar{S}$;

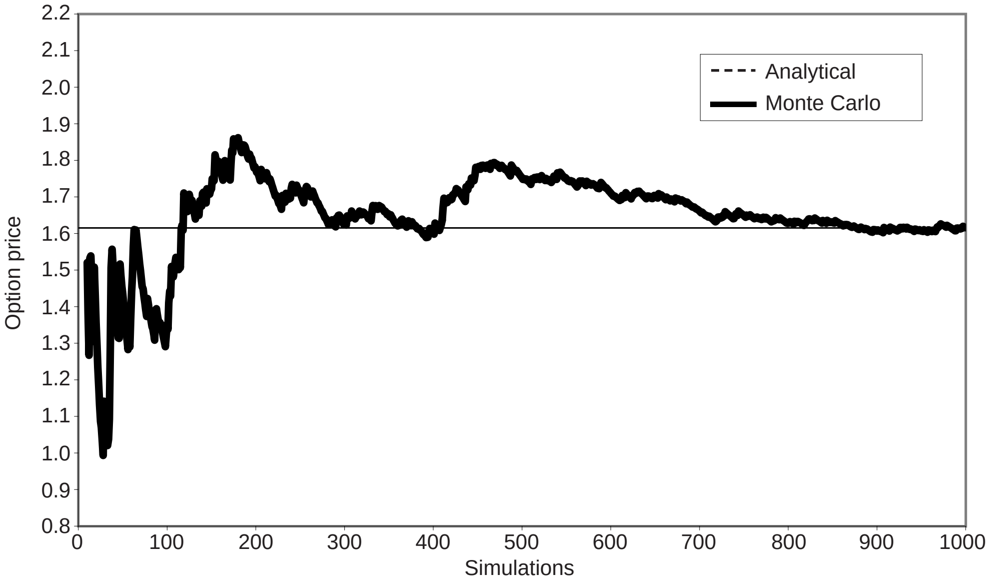

FIGURE 7.2 Convergence of Simple Monte Carlo Simulation for a European Futures Option

$$\begin{aligned}\bar{S}_i &= \exp(\bar{x}_i)\\ \bar{x}_i &= \bar{x}_{i-1} + \alpha(\hat{\mu} - \bar{x}_{i-1})\Delta t + \sigma\sqrt{\Delta t}(-\varepsilon_i)\end{aligned} \tag{7.6}$$

In other words the antithetic path is obtained by replacing ε_i by $-\varepsilon_i$ in the equation for the simulation of the asset where both processes are initialised to the same starting point, $x_0 = \bar{x}_0 = \ln S_0$. The simulated payoffs to the option under the two paths are then given for a European call futures option by;

$$C_{T,j} = \max\{0, F(T,s,S_T) - K\} \tag{7.7}$$

$$\bar{C}_{T,j} = \max\{0, F(T,s,\bar{S}_T) - K\} \tag{7.8}$$

where $F(T,s,S_T)$ denotes the s-maturity futures price observed at time T when the spot energy price is trading at S_T. We then take the average of the two payoffs as the payoff for that simulation. The antithetic method ensures that the mean of the normally distributed samples is exactly zero which also helps to improve the simulation[4]. Figure 7.3 illustrates a single simulation for valuing the same option as in figure 7.1 – the samples from the standard normal are now utilised to generate two sample paths (the

[4] Note that, not only do we obtain a more accurate estimate from n pairs of $(C_{T,j}, \bar{C}_{T,j})$ than from $2n$ evaluations of $C_{T,j}$, but it is also computationally cheaper to generate the pair $(C_{T,j}, \bar{C}_{T,j})$ than two instances of $C_{T,j}$.

S	α	μ	λ	σ	μhat		K	T	s		N
26.90	0.472	2.925	0.000	0.368	2.782		23.20	0.50	1.00		10
dt	**ln(S)**	**P(0,T)**					**call_value**		**SE**		**M**
0.05	3.292	0.951					1.628		0.052		1000

t	0	0.05	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
ε		0.708	0.574	-0.203	-0.006	-0.027	-0.013	0.367	0.629	0.434	-1.140
xt	3.292	3.338	3.373	3.342	3.328	3.313	3.300	3.318	3.357	3.379	3.271
xtbar	3.292	3.222	3.164	3.172	3.163	3.156	3.149	3.110	3.050	3.008	3.097

ST	F(T,s)	CT
26.34	24.41	1.22
22.12	21.27	0.00

FIGURE 7.3 Monte Carlo Simulation (with Antithetics) of Schwartz (1997) 1 Factor Model for a European Futures Option

rows labeled 'xt' and 'xtbar'). The payoff to the simulation is the average of the payoffs from the 2 paths, i.e. (1.22 + 0.00)/2. The Monte Carlo value is now 1.628 after 1000 simulations with a reduced standard error of 0.052.

Figure 7.4 shows the convergence of the antithetic price with 10 to 1000 simulations, as well as the convergence of the simple method repeated from figure 7.2. The Monte Carlo estimate with antithetics has visibly better convergence properties.

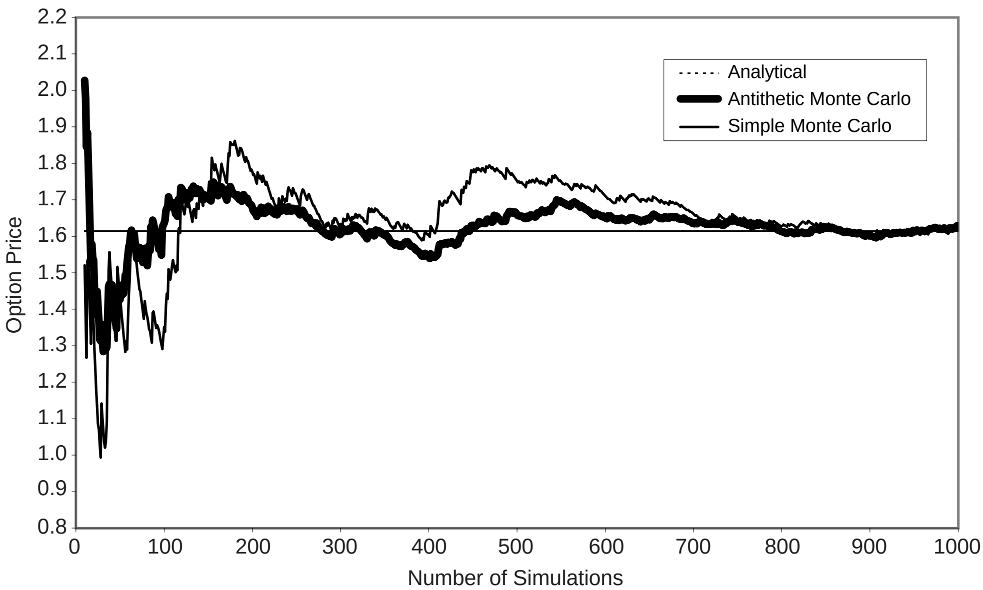

FIGURE 7.4 Convergence of Monte Carlo Simulation With Antithetics for European Futures Option

S	α	μ	λ	σ	μhat	K	T	s1	s2	N
26.90	0.472	2.925	0.000	0.368	2.782	2.04	0.50	0.75	1.50	10
dt	**ln(S)**	**P(0,T)**				**call_value**		**SE**		**M**
0.05	3.292	0.951				0.676		0.041		1000

t	0	0.05	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
ε		0.708	0.574	-0.203	-0.006	-0.027	-0.013	0.367	0.629	0.434	-1.140
xt	3.292	3.338	3.373	3.342	3.328	3.313	3.300	3.318	3.357	3.379	3.271

ST	F(T,s1)	F(T,s2)	CT
26.34	25.32	22.89	0.39

FIGURE 7.5 Monte Carlo Simulation of Schwartz (1997) 1 Factor Model for a European Calendar Spread Option

7.2.2 Example – Valuing Spread Options

The simple Monte Carlo technique from the previous section can be easily extended to model more complicated payoff conditions. For example, for the calendar spread option discussed in section 5.7.1, the user evaluates the price of the two futures contracts at the maturity of the option. Figure 7.5 illustrates a single simulation path for such as option. We assume the same process parameters as in figure 7.1 for the Schwartz model and value a 6 month option ($T = 0.5$) on the spread between futures contacts with original maturities of 9 months ($s_1 = 0.75$) and 1.5 years ($s_2 = 1.50$).

At the maturity of the option the spot price of 26.34 implies forward prices of 25.32 and 22.89 for the forward contracts with 3 months and 1 year remaining maturities respectively via equation (6.9). The option payoff of 0.39 is obtained after the evaluation of equation (5.9) assuming a strike price of 2.04. The Monte Carlo estimate after 1000 simulations is 0.676 with a standard error of 0.041.

7.2.3 Example – Valuing European Energy Swaptions

European swaptions, discussed in section 5.5, can be valued in a similar way to the spread option in the previous section. The payoff of a swaption (see equation (5.5) for a cash settled option) requires the evaluation of m forward prices at the maturity of the option in order to evaluate the swap value. Figure 7.6 shows a single simulation to price a 6 month European oil payer swaption which exercises into a 2 year semi-annual swap. The strike price of the option is assumed to be \$22.00. In figure 7.6, $\mathrm{si} = \mathrm{T} + \mathrm{i}\,\Delta\mathrm{T}$; $\mathrm{i} = 1, \ldots 4$, denote the original forward maturity dates that underlie the swap reset dates. At $t = 0.5$ the spot energy price implies forward prices at remaining maturities 0.5, 1.0, 1.5, and 2.0 of 24.41, 22.89, 21.70, and 20.76 respectively. This simulated path finishes in-the-money with an evaluated payoff of $\mathrm{CT} = 0.44$. The Monte Carlo estimate after 1000 simulations is 0.922 with a standard error of 0.056.

S	α	μ	λ	σ	μhat		K	T	ΔT		N
26.90	0.472	2.925	0.000	0.368	2.782		22.00	0.50	0.50		10
dt	**ln(S)**	**P(0,T)**					**call_value**	**SE**			**M**
0.05	3.292	0.951					0.922	0.056			1000

t	0	0.05	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
ε		0.708	0.574	-0.203	-0.006	-0.027	-0.013	0.367	0.629	0.434	-1.140
xt	3.292	3.338	3.373	3.342	3.328	3.313	3.300	3.318	3.357	3.379	3.271

ST	F(T,s1)	F(T,s2)	F(T,s3)	F(T,s4)	CT
26.34	24.41	22.89	21.70	20.76	0.44

FIGURE 7.6 Monte Carlo Simulation of Schwartz (1997) 1 Factor Model for a European Energy Swaption

7.2.4 Incorporating Seasonality into Monte Carlo Simulations

One of the most important variables which exhibits seasonal patterns is volatility. In chapter 2 we described how both a simple seasonal pattern or random volatility can be incorporated into spot price models. Random volatility is outside the scope of the models discussed in this chapter but simple seasonally varying spot volatility can easily be incorporated into the models whilst still retaining analytical tractability. In this section we show how this can be done within Monte Carlo simulation of the single factor Schwartz model, but the technique is generally applicable.

Generalising the Schwartz single factor model represented by equation (7.2) to time varying spot volatility we have;

$$dx = \alpha(\hat{\mu} - x)dt + \sigma(t)dz \tag{7.9}$$

The estimation of seasonal volatility patterns is discussed in chapters 3 and 8. Figure 7.7 illustrates a typical seasonal volatility pattern obtained from Henry Hub natural gas spot prices from 1 September 1999 to 1 March 2000.

Futures and forward prices are now given by the following equation which also relates forward prices at time t to the initial forward curve at time 0;

$$F(t,T) = F(0,T)\left(\frac{S(t)}{F(0,t)}\right)^{\exp[-\alpha(T-t)]} \exp\left[-\frac{1}{2}\int_0^t \sigma_u^2 e^{-2\alpha(T-u)}du + \frac{1}{2}e^{-\alpha(T-t)}\int_0^t \sigma_u^2 e^{-2\alpha(t-u)}du\right] \tag{7.10}$$

In general the integrals in equation (7.10) may have to be evaluated numerically but the payoff to any option which depends on forward or futures prices can be evaluated at the option maturity date as a function of the simulated spot price and parameters of the model.

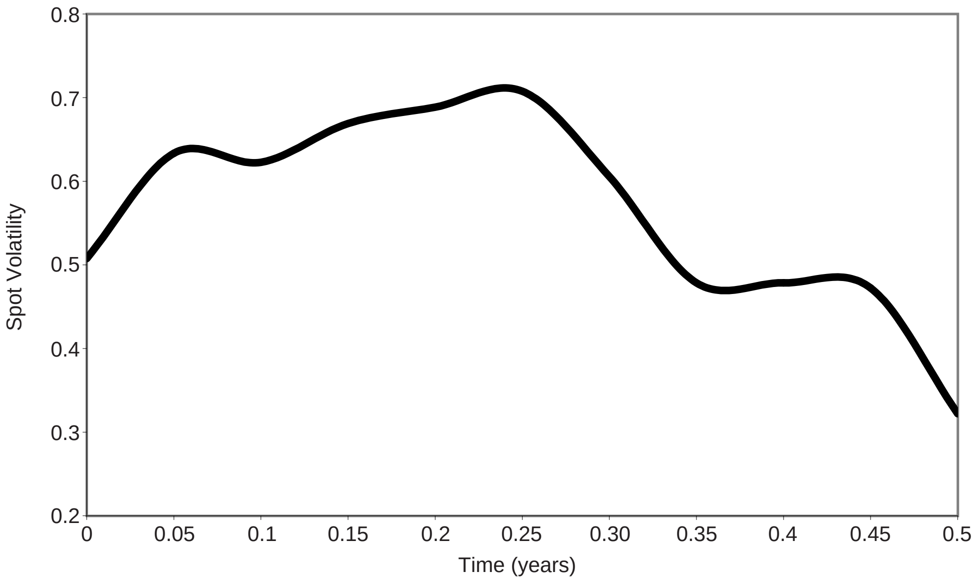

FIGURE 7.7 Typical Seasonal Energy Spot Price Volatility Pattern for Henry Hub Natural Gas

S	α	μ	λ	μhat	K	T	s	N
26.90	0.472	2.925	0.000	2.782	23.20	0.50	1.00	10
dt	**ln(S)**	**P(0,T)**			**call_value**		**SE**	**M**
0.05	3.292	0.951			2.955		0.169	1000

t	0	0.05	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
σ(t)	0.508	0.633	0.623	0.669	0.689	0.708	0.606	0.479	0.478	0.472	0.322
ε		0.708	0.574	-0.203	-0.006	-0.027	-0.013	0.367	0.629	0.434	-1.140
xt	3.292	3.361	3.428	3.385	3.370	3.352	3.336	3.373	3.426	3.457	3.321

ST	F(T,s)	CT
27.69	25.40	2.20

FIGURE 7.8 Monte Carlo Simulation of Schwartz (1997) 1 Factor Model for a European Futures Option with Seasonal Volatility

Figure 7.8 shows the same simulation as in figure 7.1 but with the seasonally varying volatility pattern shown in figure 7.7.

The simulated (log) spot path is now updated according to the time dependent spot volatility, rather than the constant value used previously.

7.2.5 Computation of Hedge Sensitivities

The standard hedge sensitivities, delta, gamma, vega, theta and rho can be computed via Monte Carlo simulation by approximating them by finite difference ratios;

$$delta = \frac{\partial C}{\partial S} \approx \frac{C(S + \Delta S) - C(S - \Delta S)}{2\Delta S} \tag{7.11}$$

$$gamma = \frac{\partial^2 C}{\partial S^2} \approx \frac{C(S + \Delta S) - 2C(S) + C(S - \Delta S)}{\Delta S^2} \tag{7.12}$$

$$vega = \frac{\partial C}{\partial \sigma} \approx \frac{C(\sigma + \Delta\sigma) - C(\sigma - \Delta\sigma)}{2\Delta\sigma} \tag{7.13}$$

$$theta = \frac{\partial C}{\partial t} \approx \frac{C(t + \Delta t) - C(t)}{\Delta t} \tag{7.14}$$

$$rho = \frac{\partial C}{\partial r} \approx \frac{C(r + \Delta r) - C(r - \Delta r)}{2\Delta r} \tag{7.15}$$

where $C(S + \Delta S)$ is the Monte Carlo estimate using an initial energy price of $S + \Delta S$, and ΔS is a small fraction of S e.g. $\Delta S = 0.001\ S$. The other $C(.)$'s are defined similarly. It is important that every price $C(.)$ should be computed using the same set of random numbers[5]. In chapter 9 we show how knowledge of these risk measures can be used to hedge the risk of energy derivative positions.

7.2.6 Simulation of Multi-Factor Models

One of the main uses of Monte Carlo simulation by industry professionals is for pricing options under multiple stochastic factors. In this section we show how the Schwartz two factor model of section (6.3) can be implemented via Monte Carlo simulation. Recall that the joint stochastic process for this model is given by;

$$dS = (r - \delta)Sdt + \sigma Sdz \tag{7.16}$$

$$d\delta = \alpha_\delta(\bar{\delta} - \delta)dt + \sigma_\delta dz_\delta \tag{7.17}$$

where the Brownian motions dz and dz_δ have instantaneous correlation $\rho_{S\delta}$.

The Monte Carlo procedure is exactly the same as that for the standard European call futures option described in the previous sections except that we simulate the joint processes given above which determine the payoff of the futures option. Equations (7.16) and (7.17) are discretised as follows;

$$\Delta S = (r - \delta)S\Delta t + \sigma S\Delta z \tag{7.18}$$

[5] The interested reader is referred to Chapter 4 of Clewlow and Strickland (1998) where the calculation of hedge sensitivities is described in detail.

$$\Delta\delta = \alpha_\delta(\bar{\delta} - \delta)\Delta t + \sigma_\delta \Delta z_\delta \tag{7.19}$$

In order to price the option by simulation we need to generate random samples of the variates Δz and Δz_δ from a standard bivariate normal distribution with correlation $\rho_{S\delta}$. This is easily achieved by generating independent standard normal variates ε_1 and ε_2 and combining them as follows[6]:

$$\Delta z = \varepsilon_1 \sqrt{\Delta t}$$
$$\Delta z_\delta = \left(\rho_{S\delta}\varepsilon_1 + \sqrt{1 - \rho_{S\delta}^2}\varepsilon_2\right)\sqrt{\Delta t}$$

As we have seen, the best way to simulate a variable following Geometric Brownian Motion is via the process for the natural logarithm of the variable which follows arithmetic Brownian motion. Therefore we have;

$$x_{i+1} = x_i + (r - \delta_i - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\varepsilon_1\sqrt{\Delta t} \tag{7.20}$$

$$\delta_{i+1} = \delta_i + \alpha_\delta(\bar{\delta} - \delta_i)\Delta t + \sigma_\delta\left(\rho_{S\delta}\varepsilon_1 + \sqrt{1 - \rho_{S\delta}^2}\varepsilon_2\right)\sqrt{\Delta t} \tag{7.21}$$

Figure 7.9 illustrates a sample simulation for crude oil prices. The process parameters are chosen to match the observed crude oil forward curve on 31 March 2000 resulting in a spot price of \$26.90, convenience yield of 26.6% and $\sigma = 36.8\%$. The convenience yield mean reverts back to a level (δbar) of 17.1% with a speed ($\alpha\delta$) of 0.363, it has a volatility ($\sigma\delta$) of

S	r	δ	σ	αδ	δbar	σδ	ρSδ	K	T	s
26.90	0.10	0.266	0.368	0.363	0.171	0.301	0.221	23.15	0.5	1.00

dt	ln(S)	N	M
0.05	3.292	10	1000

call_value	SE
2.251	0.139

t	0	0.05	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
ε1		0.691	-0.100	0.042	0.211	-0.535	-0.156	0.940	0.301	-0.522	2.464
ε2		0.879	2.244	-0.233	-0.727	0.081	-1.554	-0.860	-0.788	-0.714	-1.439
xt	3.292	3.337	3.314	3.295	3.292	3.229	3.198	3.262	3.276	3.225	3.423
δ	0.266	0.332	0.475	0.455	0.405	0.398	0.290	0.245	0.197	0.142	0.084

ST	F(T,s)	CT
30.65	30.74	7.59

FIGURE 7.9 Monte Carlo Simulation of Schwartz (1997) 2 Factor Model for a European Futures Option

[6] The general procedure for generating n correlated normal variates is described in Clewlow and Strickland (1998) where they use the eigensystem representation of the covariance matrix of the correlated variables.

30.1% and a correlation ($\rho S\delta$) of 0.221 with the spot energy price. Interest rates are assumed to be flat at 10%. The simulation is again set up to price a 6 month at-the-money European call option on a futures contract with an original maturity of 1 year with 10 time steps.

The spot price path simulation is contained in the boxed region. The first line contains the time, the second two lines contain the independent samples from a standard normal distribution, the third and fourth lines contain the simulated paths for the natural log of the spot price and the convenience yield respectively. The simulated value (call_value) after 1000 simulations is 2.251 with a standard error (SE) of 0.139. The analytical value obtained via equation (6.21) is 2.346. Notice how the price is approximately 50% greater than for the one factor model because of the additional volatility of the convenience yield which leads to the forward curve having a greater overall volatility. In this example we have not fitted the models analytical forward price volatility structure (equation (6.19)) to the actual market volatilities – this is discussed in chapter 8.

7.2.7 Generation of the Random Numbers

A critical part of Monte Carlo simulation is the generation of the standard normal random variables[7]. Clewlow and Strickland (1998) provide a discussion of the various ways to do this including the sum of twelve standard uniform random numbers, the Box-Müller transformation, and the polar rejection method. They conclude that the polar rejection method is both faster and more accurate. The method can be stated algorithmically as follows:

$$
\begin{aligned}
&\text{repeat}\\
&\quad x_1 = 2 \times \text{standard\_ uniform\_ random\_ number} - 1\\
&\quad x_2 = 2 \times \text{standard\_ uniform\_ random\_ number} - 1\\
&\quad w \;=\; x_1^2 + x_2^2\\
&\text{until } w > 1\\
&c = \sqrt{-2\frac{\ln(w)}{w}}\\
&z_1 = cx_1\\
&z_2 = cx_2
\end{aligned}
\tag{7.22}
$$

Note that the generation of x_1 and x_2 involves independent sampling of uniform random numbers. The polar rejection method generates two standard normal random variables at a time. Therefore in order to obtain maximum efficiency the second random number must be saved for use the next time a standard normal random number is required rather than generating two random numbers every time and throwing one away.

[7] The generation of the random numbers is typically about 30% of the total execution time of a Monte Carlo simulation.

Another way to make the simulations more efficient (variance reduction) is to implement quasi random numbers (also known as low discrepancy sequences). There are many ways of producing quasi random numbers and the interested reader should refer to chapter 4 of Clewlow and Strickland (1998), Joy, Boyle, and Tan (1996), Niederreiter (1992), and Niederreiter and Shiue (1995) for examples.

7.2.8 Valuing Path Dependent Options Using Simulation

In chapter 5 we discussed many so-called path dependent options – options whose payoff depends on the actual path that the energy price takes over the life of the option. In this section we look at pricing a number of popular types of path dependent option, namely barriers, lookbacks and Asians using Monte Carlo simulation.

Barrier Options

Recall from chapter 5 that there are eight basic types of barrier options: down-and-out, up-and-out, down-and-in, and up-and-in versions of both puts and calls. As the valuation principals are consistent across all eight types of options we limit our discussion here to pricing down-and-out calls on the forward energy price. We value a European down and out call option, maturing at time T, which expires worthless if the underlying forward price, $F(t,s)$, crosses the barrier on certain predefined reset dates. The payoff can be represented by[8];

$$\max(0, F(T,s) - K)1_{\min(F(t_1,s),\ldots,F(t_m,s))>H} \tag{7.23}$$

Figure 7.10 shows a sample simulation. We value a down-and-out call option with the

S	α	μ	λ	σ	μhat	K	T	s	H	N
26.90	0.472	2.925	0.000	0.368	2.782	23.20	0.50	1.00	22.04	10

dt	ln(S)	P(0,T)	F(0,s)	call_value	SE	M
0.05	3.292	0.951	23.20	1.264	0.083	1000

t	0	0.05	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
ε		0.417	-0.877	2.114	-0.969	-0.920	1.609	-0.644	-2.190	1.049	0.799
xt	3.292	3.314	3.230	3.393	3.299	3.211	3.333	3.267	3.075	3.155	3.212
F(t,s)	23.20	23.68	22.55	25.30	23.91	22.64	24.85	23.86	**20.78**	22.17	23.30

CT
0.00

FIGURE 7.10 Monte Carlo Simulation of Schwartz (1997) 1 Factor Model for a European Barrier Option

[8] The forward price which is checked against the barrier may be different to the forward price determining the payoff to the option.

same process characteristics as the option of figure 7.1 but with a knock-out barrier contingent on the 1 year forward price. The 1 year forward price is observed[9] to be \$23.20 and the barrier is set to be 5% below the current forward underlying price ($H =$ \$22.04). The option is designed to knock out if the underlying forward crosses the barrier. Although the simulated path finishes in-the-money for a standard call option, the forward price at time 0.4 trades at a level of \$20.78 which is below the barrier level and so the option expires worthless. Simulating 1000 payoffs yields a Monte Carlo estimate of the price of 1.264 with a standard error of 0.083 for this particular option. Note that the value of the option is significantly less than the standard option priced in figure 7.1 because of the possibility of the option knocking out.

Lookback Options

In chapter 5 it was shown that the terminal payoff to a lookback option is a function of the highest or lowest underlying energy price during the lifetime of the option and that there are four main types which are summarised in table 5.3. As an example we take a floating strike lookback call option with maturity $T = t_m$ where the payoff is determined by the minimum level of the underlying forward price, $F(t,s)$, over the life of the option[10];

$$\max(0, F(T,s) - \min(F(t_1,s), ..., F(t_m,s))) \tag{7.24}$$

Figure 7.11 shows a sample simulation for pricing a 6 month floating strike lookback call option where the option is fixed on the levels of the forward contract with 1 year to maturity at the initial time. The process parameters and fixings are as described in figure 7.1.

S	**α**	**μ**	**λ**	**σ**	**μhat**			**T**	**s**		**N**
26.90	0.472	2.925	0.000	0.368	2.782			0.50	1.00		10
dt	**ln(S)**	**P(0,T)**	**F(0,s)**				**call_value**		**SE**		**M**
0.05	3.292	0.951	23.20				2.495		0.089		1000

t	**0**	**0.05**	**0.1**	**0.15**	**0.2**	**0.25**	**0.3**	**0.35**	**0.4**	**0.45**	**0.5**
ε		-0.200	-0.519	0.503	1.427	-1.359	0.164	-1.619	-0.993	0.266	0.246
xt	3.292	3.264	3.210	3.241	3.348	3.222	3.225	3.082	2.993	3.010	3.025
F(t,s)	23.20	22.92	22.26	22.85	24.72	22.82	23.00	20.81	**19.53**	19.82	20.10

CT
0.57

FIGURE 7.11 Monte Carlo Simulation of Schwartz (1997) 1 Factor Model for a European Lookback Option

[9] The forward price has been calculated via equation (6.9).
[10] The minimum could also be determined over a partial period of the option life.

The minimum price of the forward contract occurs at time $t = 0.4$ under the simulation in figure 7.11 at 19.53. The payoff under the simulation is therefore equal to max(0, 20.10–19.53) = 0.57. With 1000 simulations the Monte Carlo estimate of the price is 2.495 with a standard error of 0.089. The value of this lookback is much greater than the equivalent standard call option valued in figure 7.1 because this option effectively allows the holder to buy at the minmum price.

Asian Options

Finally in this section we look at options based on the average of energy prices - Asian options. As an example we price a European fixed strike Asian call option with maturity T, where the averaging is applied to the price of a forward contract observed 10 times during the life of the option[11]. Recall from chapter 5 that the payoff for the option is determined as the difference between the average of the observed forward energy prices at the fixing times t_k and the strike price K;

$$\max(0, \frac{1}{m}\sum_{k=1}^{m} F(t_k, s) - K) \tag{7.25}$$

Figure 7.12 shows a sample simulation for the pricing of such an option. We value a 6 month option on the price of a forward contract with an initial maturity of 1 year. The average of the forward prices for the simulation in figure 7.12 is 24.60. The payoff to the Asian option under this simulation is therefore max(0, 24.60 − 23.20) = 1.40. Running 1000 simulations gives a Monte Carlo estimate of 0.891 with a standard error of 0.048. The Asian option is worth approximately half of the equivalent standard call option in

S	α	μ	λ	σ	μhat	K	T	s	N
26.90	0.472	2.925	0.000	0.368	2.782	23.20	0.50	1.00	10

dt	ln(S)	P(0,T)	F(0,s)	call_value	SE	M
0.05	3.292	0.951	23.20	0.891	0.048	1000

t	0	0.05	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
ε		0.708	0.574	-0.203	-0.006	-0.027	-0.013	0.367	0.629	0.434	-1.140
xt	3.292	3.338	3.373	3.342	3.328	3.313	3.300	3.318	3.357	3.379	3.271
F(t,s)	23.199	24.045	24.762	24.448	24.398	24.319	24.256	24.755	25.688	26.351	24.414

CT
1.40

FIGURE 7.12 Monte Carlo Simulation of Schwartz (1997) 1 Factor Model for a European Asian Call Option

[11] Partial averaging periods are also popular.

figure 7.1 because the uncertainty in the average is much less than that of the underlying forward price at the option maturity.

7.3 Building Trinomial Trees For the Energy Spot Process

In this section we describe a general and efficient procedure involving the use of trinomial trees for modelling the spot energy price so that it is consistent with initial futures market data[12] and allows the pricing of American and path-dependent options. This method can also be easily extended to incorporate seasonal volatility and/or volatility smiles and skews, extensive details are given in chapter 5 of Clewlow and Strickland (1998)[13]. The 'fitting' of the model to the market prices of futures/forwards is achieved by introducing a time dependent mean reversion level into the drift of the spot price. Clewlow and Strickland (1999a) (CS) develop a procedure for building trinomial trees to achieve this consistency between the model and the market. These trees can then be used for pricing American options as well as options with path dependent characteristics. American option valuation requires evaluation of the following expression;

$$C(t) = \underset{\theta \in \Psi[t,T]}{Max} \mathrm{E}_t \left[\exp\left(-\int_t^\theta r(u)du \right) C^*(\theta) \right] \tag{7.26}$$

where $C^*(\theta)$ is the payoff of the option when it is exercised at date θ and $\Psi[t, T]$ is the class of all early exercise strategies (stopping times) in $[t,T]$. The early exercise strategy, and hence the option price, can be easily determined from the tree for the spot energy price.

Defining x as the natural logarithm of the energy spot price, $x(t) = \ln(S(t))$, CS show that to achieve consistency with the observed forward curve for the Schwartz single factor model the process for x is given by;

$$dx(t) = [\theta(t) - \alpha x(t)]dt + \sigma dz(t) \tag{7.27}$$

$$\theta(t) = \left(\frac{\partial \ln F(0,t)}{\partial t} + \alpha \ln F(0,t) + \frac{\sigma^2}{4}\left(1 - e^{-2\alpha t}\right) - \frac{1}{2}\sigma^2 \right)$$

The addition of a time dependent parameter in the drift term allows the model to return the market futures prices. The CS tree building procedure consists of two stages. First, a preliminary tree is built for x assuming that $\theta(t) = 0$ and the initial value of x is zero. The resulting 'simplified' process for this new variable, $\bar{x}$, is given by:

$$d\bar{x}(t) = -\alpha \bar{x}(t)dt + \sigma dz(t) \tag{7.28}$$

The time values considered by the tree are equally spaced and have the form $i\Delta t$ where i

[12] This section is based on the paper "Valuing Energy Options in a One Factor Model Fitted to Forward Prices" by Clewlow and Strickland (1999a). See also Clewlow and Strickland (1999d).

[13] Trees can also be extended to incorporate jumps, both in the spot price and volatility, but this is beyond the computational scope of this book. The interested reader is referred to Amin (1994) and Naik (1994).

is a non-negative integer. A reasonable choice for the relationship between the space step and the time step is given by:

$$\Delta\bar{x} = \sigma\sqrt{3\Delta t} \tag{7.29}$$

Assume that the value of $\bar{x}$ at the j-th node at time t_i is $\bar{x}_{i,j}$ where j determines the level of the variable in the tree. The trinomial branching process and the associated probabilities are chosen to be consistent with the drift and volatility of the process (7.28). The three nodes which can be reached by the branches emanating from node (i,j) are $(i+1, k-1)$, $(i+1, k)$, and $(i+1, k+1)$ where k is chosen so that the value of $\bar{x}$ reached by the middle branch is as close as possible to the expected value of $\bar{x}$ at time t_{i+1} . The expected value of $\bar{x}$ at time t_{i+1} conditional on $\bar{x} = \bar{x}_{i,j}$ is $\bar{x}_{i,j} - \alpha\bar{x}_{i,j}\Delta t$.

Let $p_{u,i,j}$, $p_{m,i,j}$, and $p_{d,i,j}$ define the probabilities associated with the lower, middle, and upper branches emanating from node (i,j) respectively. These probabilities are given by:

$$\begin{aligned}
p_{u,i,j} &= \frac{1}{2}\left[\frac{\sigma^2\Delta t + \alpha^2 x_{i,j}^2 \Delta t^2}{\Delta x^2} + (k-j)^2 - \frac{\alpha x_{i,j}\Delta t}{\Delta x}(1+2(k-j)) + (k-j)\right] \\
p_{d,i,j} &= \frac{1}{2}\left[\frac{\sigma^2\Delta t + \alpha^2 x_{i,j}^2 \Delta t^2}{\Delta x^2} + (k-j)^2 + \frac{\alpha x_{i,j}\Delta t}{\Delta x}(1+2(k-j)) + (k-j)\right] \qquad (7.30) \\
p_{m,i,j} &= 1 - p_{u,i,j} - p_{d,i,j}
\end{aligned}$$

The procedure described so far applies to the process $\bar{x}$ with $\theta(t) = 0$ and $\bar{x} = 0$.

The second stage in the tree building procedure consists of displacing the nodes in the first tree in order to add the proper drift and to be consistent with the observed term structure. We can introduce the correct time varying drift, by displacing the nodes at time $i\Delta t$ by an amount a_i which are chosen to ensure that the tree correctly returns the observed forward price curve. The value of x at node (i,j) in the new tree equals the value of $\bar{x}$ at the corresponding node in the original tree plus a_i; the probabilities remain unchanged. The key to this stage is the use of forward induction to ensure that all forward prices are priced correctly.

In order to determine these time dependent functions we can use pure security (or state or Arrow-Debreu) prices. Defining $Q_{i,j}$ as the value, at time 0, of a security that pays \$1 if node (i,j) is reached, or \$0 otherwise, we see that Arrow-Debreu securities are the building blocks of all securities; in particular the price today, $C(0)$ of any European claim with payoff function $C(S)$ at time step i in the tree is given by;

$$C(0) = \sum\nolimits_j Q_{i,j} C(S_{i,j}) \tag{7.31}$$

where the summation takes place across all of the nodes j at time i. The state prices are accumulated according to the relationship:

$$Q_{i+1,j} = \sum_{j'} Q_{i,j'} p_{j',j} P(i\Delta t, (i+1)\Delta t) \tag{7.32}$$

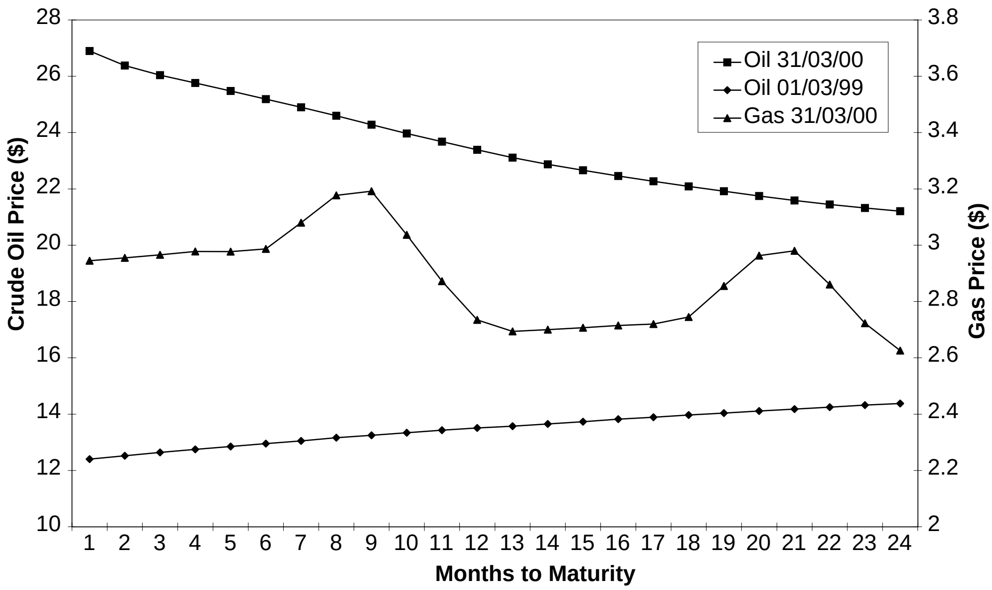

FIGURE 7.13 Typical Energy Market Forward Curves

where $p_{j',j}$ is the probability of moving from node $(i,\ j')$ to node $(i+1,\ j)$ and $P(i\Delta t, (i+1)\Delta t)$ denotes the price at time $i\Delta t$ of the pure discount bond maturing at time $(i+1)\Delta t$. In order to use the state prices to match the forward curve we use the following special case of equation (7.31);

$$P(0, i\Delta t)F(0, i\Delta t) = \sum_j Q_{i,j}S_{i,j} \tag{7.33}$$

CS show that the adjustment term needed to ensure that the tree correctly returns the observed futures curve is given explicitly as;

$$a_i = \ln\left(\frac{P(0, i\Delta t)F(0, i\Delta t)}{\sum_j Q_{i,j}e^{\tilde{x}_{i,j}}}\right) \tag{7.34}$$

To illustrate the procedure we have fitted the spot price tree to a number of different market forward curves. Figure 7.13 shows three market curves that are representative of; a downward sloping forward price curve (NYMEX Light, Sweet Crude Oil Futures Contracts, 31 March 2000), an upward sloping curve (NYMEX Light, Sweet Crude Oil Futures Contracts, 31 March 1999), and a forward curve which exhibits seasonality (NYMEX Henry Hub Natural Gas Futures Contracts, 31 March 2000). Two years worth of monthly maturity contracts are used to construct the curves.

Figure 7.14 shows the resulting trees with time steps every two months.

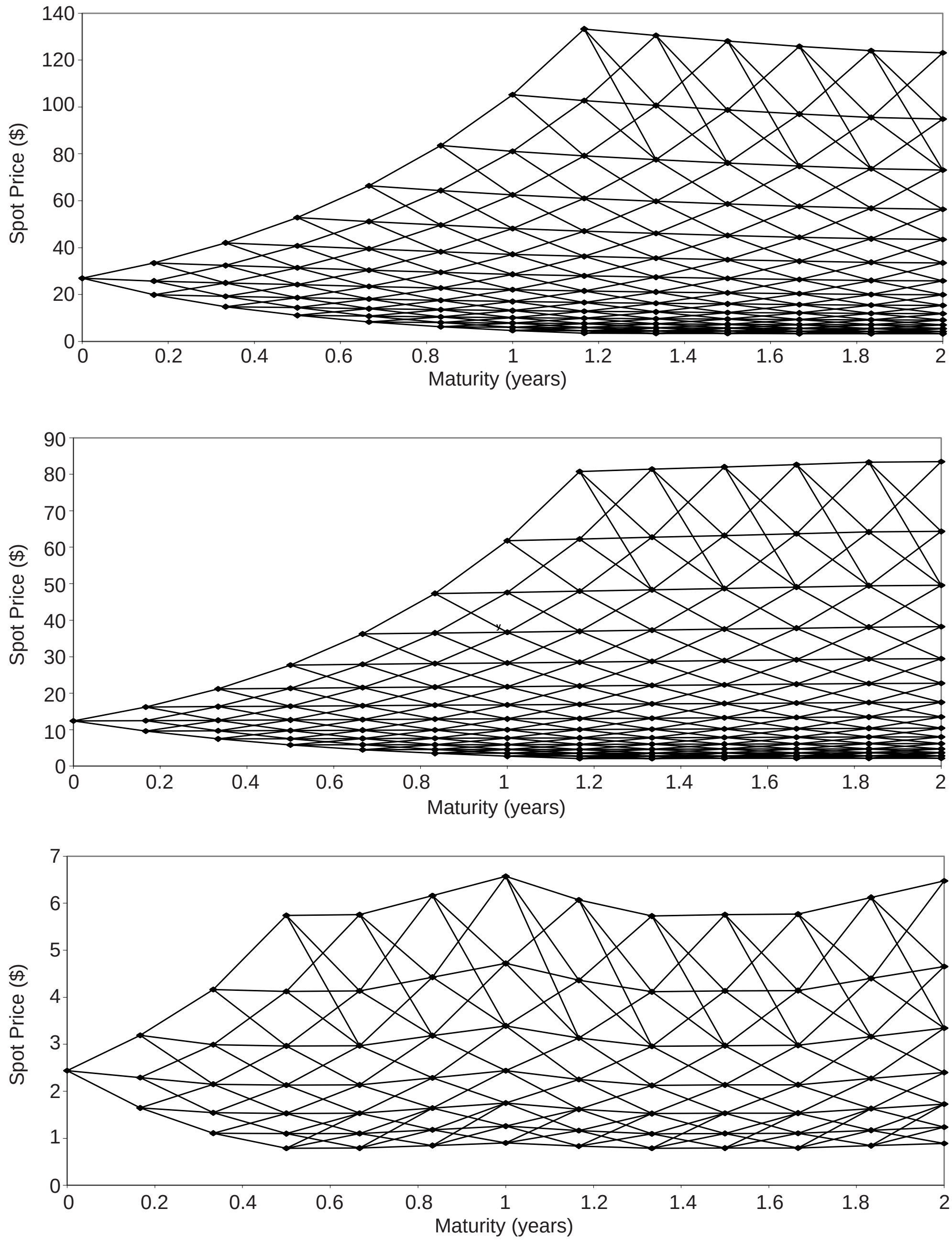

FIGURE 7.14 Spot Price Trees Fitted to Market Forward Curves (Downward sloping, Upward Sloping, and Seasonal)

TABLE 7.1 Value of European and American Options Calculated From the Tree

	Options on Spot				Options on Future			
Steps/ Year	Euro Call	Euro Put	Amer Call	Amer Put	Euro Call	Euro Put	Amer Call	Amer Put
50	2.482	2.482	4.320	2.596	1.408	2.484	1.429	2.542
100	2.470	2.470	4.322	2.586	1.413	2.490	1.435	2.547
150	2.465	2.465	4.320	2.580	1.406	2.483	1.428	2.540
200	2.461	2.461	4.319	2.577	1.409	2.486	1.431	2.543
250	2.463	2.463	4.321	2.579	1.410	2.487	1.432	2.544
300	2.464	2.464	4.322	2.580	1.409	2.485	1.431	2.543
350	2.465	2.465	4.321	2.581	1.406	2.483	1.429	2.541
400	2.463	2.463	4.321	2.581	1.408	2.484	1.429	2.541
Analytical	2.463	2.463	N/A	N/A	1.408	2.484	N/A	N/A

The volatility parameters used in the tree construction were chosen by best fitting, in a least squares sense, the negative exponential forward price volatility function to sample standard deviations of one years worth of historical daily futures returns. The resulting parameters for the speed of mean reversion and spot price volatility were 0.472 and 0.368 respectively for crude oil, and 1.436 and 0.468 for the gas data.

Table 7.1 shows the results of pricing a one year at-the-money (forward) option on crude oil. The tree was constructed to fit the downward sloping forward curve of crude oil on the 31 March 2000 from figure 7.13. Prices for European and American exercise style options on the spot and a 1.5 year forward contract are determined from the tree for different numbers of time steps. The volatility parameters used in the tree construction were chosen by a best fit to sample standard deviations for one year of historical data prior to 31 March 2000. Interest rates are assumed to be 10%.

We also compare the prices of European options calculated from the tree with the analytical values calculated via equations (6.12) and (6.14). Table 7.1 illustrates that prices calculated from the tree converge to the analytical price. It can also be seen from table 7.1 that there is an early exercise premium associated with both options on the spot price and on the forward price due to the fact that the downward sloping forward curve implies a significant convenience yield on the spot asset.

The nature of the construction of the tree implies that hedge parameters can be quickly and easily calculated. If we calculate hedge parameters with respect to some 'shift' in the forward curve, then this shift only affects the displacement coefficients – it doesn't effect the position of the branches relative to the central branch or the probabilities associated with the branches.

7.4 Implied Trinomial Trees

In a number of earlier chapters we have discussed the volatility smiles commonly found in traded options prices, which contain important information about the markets expectation of the future. A key issue for many options traders is how to incorporate this volatility information into numerical schemes in order to be able to value exotic options (the subject of the next section) consistent with the information on the smile embedded in standard option prices. One technique is to build implied trees. Implied trees can be considered as generalisations of the trinomial trees we have so far described. This generalization can be achieved by making previously constant parameters (such as the probabilities) time dependent and to imply these time dependent parameters such that the tree returns the market prices of the standard options – in this way the structure of the implied trinomial tree will be very similar to that of the constant coefficient trinomial tree. A detailed description of the construction of trinomial trees is beyond the scope of this chapter, but the interested reader is referred to chapter 5 of Clewlow and Strickland (1998) where implied trees are discussed at great length.

7.5 Pricing General Path Dependent Energy Options in Spot Price Trees[14]

Having constructed trinomial trees for the spot energy process in the previous section, we outline in this section how to price general path dependent options using the techniques developed in Hull and White (1993) (HW) for a Black-Schole-Merton world and extended by Clewlow and Strickland (1997) (CS) which contains further details and examples of this technique.

Assume we wish to price a general path dependent option whose payoff depends on some function $G(F(t,s); 0 \leq t \leq T, t \leq s)$ of the path of the forward price curve. The procedure developed in HW and CS follows a number of steps. Firstly, the user determines the range (i.e. the minimum and maximum) of the possible values of $G(.)$ which can occur for every node in the tree. This is achieved by stepping forward through the tree from the origin to the maturity date computing, at each node, the minimum and maximum value of $G(.)$ given the value at the nodes at the previous time step which have branches to the current node and the forward curve at the current node.

Secondly, we choose an appropriate set of values of $G(.)$ between the minimum and maximum possible for each node. In choosing this set of values we note that the nodes which lie on the upper and lower edges of the tree have only one path which reaches them and therefore there can be only one value of $G(.)$. The largest range of values will typically occur in the central section of the tree. The number of values we consider should in general increase only linearly with the number of time steps and also decrease linearly from the central nodes of the tree down to one at the edges of the tree in order to control the computational requirements. Let $n_{i,j}$ be the number of values we store at node (i,j) and $G_{i,j,k}$, $k = 1, \ldots, n_{i,j}$ be the values of $G(.)$ where $G_{i,j,1}$ is the minimum and $G_{i,j,n_{i,j}}$ is the maximum. Clewlow and Strickland (1998) suggest choosing $n_{i,j}$ to be

$$n_{i,j} = 1 + \beta(i - abs(j)) \tag{7.35}$$

[14] This section is based on section 5 of Clewlow and Strickland (1999a).

so that $n_{i,j}$ will always be one at the edges of the tree and $1+\beta i$ in the center of the tree. In this way we can increase β to increase the accuracy of the approximation by considering more values of $G(.)$. In choosing the actual set of $n_{i,j}$ values for each node we should consider the distributional properties of the function $G(.)$. This will vary depending on the nature of $G(.)$ and therefore must be considered on a case by case basis.

The third step in the procedure is to set the value of the option at maturity at every node and for every value of $G(.)$

$$C_{N,j,k} = C(t_N, G_{N,j,k});\ \forall j,k \tag{7.36}$$

Finally, we step back through the tree computing discounted expectations and applying the early exercise condition at every node and for every value of $G(.)$

$$C_{i,j,k} = e^{-f(i,i+1)\Delta t}\left(p_{u,i,j}C_{i+1,j+1,u} + p_{m,i,j}C_{i+1,j,m} + p_{d,i,j}C_{i+1,j-1,d}\right) \tag{7.37}$$

where $f(i,i+1)$ denotes the one period forward rate from time step i to time step $i+1$ and where $C_{i+1,j+1,u}$, $C_{i+1,j,m}$, $C_{i+1,j-1,d}$ are the values of the option at time step $i+1$, given the current $G_{i,j,k}$, for upward, middle and downward branches of the spot price. These are obtained by computing the value of $G(.)$, given the current value, after upward, middle and downward branches $G_{i+1,j+1,u}$, $G_{i+1,j,m}$, $G_{i+1,j-1,d}$.

The values $G_{i+1,j+1,u}$, $G_{i+1,j,m}$, $G_{i+1,j-1,d}$, and therefore also the option values $C_{i+1,j+1,u}$, $C_{i+1,j,m}$, $C_{i+1,j-1,d}$, will not in general be stored at the upward, middle and downward nodes and therefore must be obtained by interpolation. For example using linear interpolation we have

$$C_{i+1,j+1,u} = C_{i+1,j+1,k_l} + \left(\frac{C_{i+1,j+1,k_u} - C_{i+1,j+1,k_l}}{G_{i+1,j+1,k_u} - G_{i+1,j+1,k_l}}\right)\left(G_{i+1,j+1,u} - G_{i+1,j+1,k_l}\right) \tag{7.38}$$

where k_l and k_u are such that $G_{i+1,j+1,k_l} \le G_{i+1,j+1,u} \le G_{i+1,j+1,k_u}$ and $k_u = k_l + 1$. That is the two values of $G(.)$ which lie closest to either side of $G_{i+1,j+1,u}$ are found and a linear interpolation between these is done to obtain an estimate for $C_{i+1,j+1,u}$. The value of the path dependent contingent claim is read from the tree as the value of $C_{0,0,0}$.

7.5.1 Pricing Asian Energy Options in a Trinomial Tree

As a specific example of the generalised methodology outlined in the previous section we price European and American versions of an average price call option, where the average is taken over the spot energy price on the fixing dates t_l, $l = 1, \ldots, L$.

Let there be a total of N time steps from the start of the life of the option until its maturity. In order to find the range of values of the average at each node we step forward through the tree from $i = 0$ to $i = N$. If we have found the range for all nodes up to time step $i-1$ then for any node (i,j) the minimum average is determined by the minimum average of the lowest node at time step $i-1$ with a branch to the current node and the spot price at the current node. The minimum average is given by

$$G_{i,j,1} = \begin{cases} \frac{G_{i-1,j_l,1} m_{i-1} + S_j}{m_i} & \text{if } t_i = t_{m_i} \text{ i.e. a fixing date} \\ G_{i-1,j_l,1} & \text{otherwise} \end{cases} \tag{7.39}$$

where m_i is the number of fixing dates which have occurred up to time step i and node $(i-1, j_l)$ is the lowest node with a branch to node (i,j). Similarly the maximum average is determined by the maximum average of the highest node at time step $i-1$ with a branch to the current node and the energy spot price at the current node

$$G_{i,j,n} = \begin{cases} \frac{G_{i-1,j_u,n} m_{i-1} + S_j}{m_i} & \text{if } t_i = t_{m_i} \text{ i.e. a fixing date} \\ G_{i-1,j_u,n} & \text{otherwise} \end{cases} \tag{7.40}$$

where node $(i-1, j_u)$ is the highest node with a branch to node (i,j). Now since the arithmetic average of the spot price is essentially a sum of lognormally distributed prices it will also be approximately lognormally distributed. We therefore choose a log-linear set for the $n_{i,j}$ values of the average at each node (i,j) which gives

$$G_{i,j,k} = G_{i,j,1} e^{(k-1)h} \tag{7.41}$$
$$h = \frac{\ln(G_{i,j,n}) - \ln(G_{i,j,1})}{n_{i,j} - 1}$$

In order to determine the option values of equation (7.37) we first compute what the average would be, given the current average, after upward, middle and downward branches $G_{i+1,j+1,u}$, $G_{i+1,j,m}$, $G_{i+1,j-1,d}$

$$G_{i+1,j+1,u} = \begin{cases} \frac{G_{i,j,k} m_i + S_{j+1}}{m_{i+1}} & \text{if } t_{i+1} = t_{m_{i+1}} \text{ i.e. a fixing date} \\ G_{i,j,k} & \text{otherwise} \end{cases} \tag{7.42}$$

$$G_{i+1,j,m} = \begin{cases} \frac{G_{i,j,k} m_i + S_j}{m_{i+1}} & \text{if } t_{i+1} = t_{m_{i+1}} \text{ i.e. a fixing date} \\ G_{i,j,k} & \text{otherwise} \end{cases} \tag{7.43}$$

$$G_{i+1,j-1,d} = \begin{cases} \frac{G_{i,j,k} m_i + S_{j-1}}{m_{i+1}} & \text{if } t_{i+1} = t_{m_{i+1}} \text{ i.e. a fixing date} \\ G_{i,j,k} & \text{otherwise} \end{cases} \tag{7.44}$$

As an example we price European and American versions of a fixed strike average price call option on crude oil with 1 year to maturity and where the terminal payoff is determined by the daily average of the crude oil price during the last month of the life of the option. The valuation date is the 31 March 2000, the tree is constructed to be consistent with the downward forward curve in Figure 7.7, using the same parameters as used for table 7.1. Table 7.2 contains the results.

Table 7.2 shows the convergence of both the European and American option values as we increase both the numbers of time steps per year and also the number of averages at each node.

TABLE 7.2 Convergence of European and American Fixed Strike Average Price Call Options

	Number of Averages at Each Node of Tree ($n_{i,j}$)					
	4		12		20	
Steps/Year	Euro	Amer	Euro	Amer	Euro	Amer
50	2.526	2.702	2.488	2.671	2.484	2.667
100	2.575	2.789	2.491	2.711	2.484	2.704
150	2.629	2.873	2.499	2.737	2.486	2.726
200	2.684	2.953	2.507	2.755	2.488	2.738
250	2.738	3.033	2.520	2.773	2.490	2.746
300	2.792	3.110	2.534	2.793	2.495	2.753
350	2.874	3.225	2.559	2.830	2.505	2.771
400	2.926	3.297	2.575	2.852	2.513	2.780

7.5.2 Pricing Swing Options

The term swing option refers to flexibility in the quantity of energy which the holder of the option can receive. The concept being that the volume of energy can 'swing' between specified limits. Swing options have arisen because of the uncertainty in the volume of energy, for example gas or electricity, that the end user will consume. Energy suppliers are exposed to this volume risk and swing options were created to allow this risk to be at least partially hedged.

Swing options can be thought of as a series of flexible forward contracts or swaps. In general the specification of a swing option will involve the total time period of the contract, the trade dates within that period on which the energy can be bought (or sold) and the purchase (or selling) price which may be fixed or floating. The contract will also specify the limits on the quantity of energy involved – typically a minimum and maximum volume per trade date and an overall minimum and maximum for the contract.

The difficulty in pricing swing options is that the optimal decision on the quantity of energy to buy (or sell) on each date depends not only on the energy price but also on the quantity of energy already bought (or sold). Thus swing options can be characterised as path dependent American style options. The approach required to price these type of contracts is similar to that described in section 7.4 in which the quantity of energy becomes the path dependent variable.

To illustrate the pricing approach for swing options we use the example of the 'Take-or-Pay' option (ToP). The ToP contract specifies a series of dates, t_i; $i = 1, \ldots, m$ and prices, K_i, at which the underlying energy can be bought together with a maximum quantity, $V_{\max}$, on each date and an overall minimum quantity, $V_{\min}$, which must be purchased in order to avoid a penalty, q. The dates are typically spread over a one year period and the penalty is usually a specified fraction of the final date purchase price applied to the volume deficit. On each date the decision involves a trade-off between the profit or loss which would be achieved by purchasing energy at the specified contract price relative to the actual market price and the need to purchase sufficient energy overall to minimise the penalty payment at the energy of the contract.

Assume we have built a tree consistent with the spot energy price process where i indicates the time step and j indicates the spot price level as described in section 7.3. The first step is to determine the range of values of the path dependent volume of energy purchased. In fact it is more convenient to choose the path dependent variable to be the volume of energy which remains to be purchased in order to avoid the penalty. The range of values which this can take is from zero (if enough energy has already been purchased to avoid the penalty) up to the minimum required to avoid the penalty (if no energy has been purchased). Next we choose a discrete set of values for the volume of energy between the minimum and maximum, $V_{i,j,k}$; $k = 1, \ldots, n_{i,j}$, the greater the number of values, $n_{i,j}$, the more accurate will be the computed price of the contract.

The value of the contract at maturity, $i = N$, (the final purchase date) can then be computed. The final decision is relatively simple because the penalty amount is known with certainty. If the spot price is greater than the purchase price, K_m, then the payoff is strictly increasing in the volume purchased and the maximum quantity of energy, $V_{\max}$, should be purchased. If the spot price is less than K_m but greater than $(1-z)\ K_m$ then purchasing a quantity up to that required to avoid the penalty or the maximum possible, whichever is the smaller is optimal. This is because the loss on the purchase of the energy is more than compensated by the reduction in the penalty payment. If the spot price is less than $(1-q)\ K_m$ then no purchase of energy is the optimal decision. The optimal volume to take, $V^*_{N,j,k}$, and the value of the contract, $C^*_{N,j,k}$, can be summarised as follows:

$$V^*_{N,j,k} = \begin{cases} V_{\max} & \text{for} \quad S_{N,j,k} > K_N \\ \min(V_{N,j,k}, V_{\max}) & \text{for} \quad K_N > S_{N,j,k} \geq (1-z)K_N \\ 0 & \text{for} \quad (1-z)K_N > S_{N,j,k} \geq 0 \end{cases} \tag{7.45}$$

$$C^*_{N,j,k} = (S_{N,j,k} - K_N)V^*_{N,j,k} - zK_N \max(0, V_{N,j,k} - V^*_{N,j,k}) \tag{7.46}$$

Finally, we step back through the tree computing the discounted expectations of the contract value at each node and computing the optimal purchase decision at the purchase dates. The optimal purchase decision at each node and for each value of the remaining volume can be computed by searching over the range of possible purchase volumes (zero to $V_{\max}$) for the volume which maximises the sum of the discounted expectation[15] and the value of the current purchase. At the root node ($i = 0, j = 0$) the

[15] The discounted expectation depends on the current volume purchase decision, since this affects the volume remaining to be purchased in the future to avoid the penalty payment.

volume remaining to be purchased to avoid the penalty is known and therefore the value of contract only needs to be computed for the associated value of k. For a new contract the volume remaining will be $V_{\min}$, whereas for an existing contract this may not be the case.

Take-or-Pay contracts often have an added complexity, referred to as a make-up provision. These type of ToP contracts are usually multi-year or have multiple penalty dates. The penalty paid on each penalty date can be credited against future purchases above the minimum volume to be purchased in the following year. This extra flexibility introduces another path dependent state variable which can be captured by the volume remaining which can have a penalty credit applied to it. The solution procedure for this more complex problem is basically the same as for the simple case, however, at each node and for each pair of values of the path dependent variables the optimal decision must be found by searching a two-dimensional grid of the variables. Solving these types of problems is therefore correspondingly more computationally demanding.

7.6 Summary

In this chapter we have discussed two of the most popular techniques used by practitioners for pricing energy derivatives using spot price models, when there are no analytically tractable solutions. Monte Carlo simulation is shown to be a flexible technique for pricing many types of path dependent option, which can easily be extended to handle seasonality and to implement multi-factor models. We have illustrated the technique with examples showing the pricing of futures options, spread options, and energy swaptions, as well as a number of path dependent options including barrier, lookback and Asian options. For pricing options that have early exercise opportunities we have shown how to construct a trinomial tree for a mean-reverting spot model fitted to the observed forward curve. Although the implementation of this technique is more difficult than simulation, it allows the pricing of American style options in addition to European pricing. Path dependent options, including swing contracts, can also be handled relatively straightforwardly in this framework.